

PRISM — Polarimetric Resource Intelligence System for the Moon

Complete Project Outline v2.0 — PS 8 (Hackathon Edition)

Detection and Characterization of Subsurface Ice in Lunar South Polar Regions

SECTION 0 — CRITICAL LOOPHOLES IN v1 OUTLINE (FIXED HERE)

Before anything else, here is an honest audit of what was weak or broken in the previous outline:

Loophole 1 — No fallback if DFSAR SLC data is not full-polarimetric The original outline assumes DFSAR data will be full-polarimetric (quad-pol), enabling Yamaguchi decomposition. DFSAR operates in both full-pol and compact-pol modes — if the supplied data is compact-pol, the covariance matrix cannot be built the same way. Fix: always check data mode first. For compact-pol data, use m-chi or m-delta decomposition (Raney 2012) instead of Yamaguchi — the pipeline still works, just with different decomposition algebra.

Loophole 2 — Random Forest training labels are circular Block 2 proposes using "high-confidence CPR anomaly pixels from the PRL paper" as positive training examples, then evaluating the RF on CPR maps. The labels and features overlap — this is label leakage and will produce inflated accuracy. Fix: positive labels must come from an independent source — use Mini-RF (LRO) confirmed ice pixels, or use the PRL paper's geographic polygon boundaries as training regions only, not CPR values themselves. Evaluate on held-out craters (F3, H3), not F2.

Loophole 3 — Depth stratification uses a hard threshold, not uncertainty-aware The 3-class depth map (shallow / deep / none) uses hard CPR thresholds with no confidence interval. A pixel at CPR = 1.01 and one at CPR = 2.5 get the same label. Fix: propagate CPR uncertainty (speckle noise → ENL-derived variance) into the depth classification. Output per-pixel class probabilities, not hard labels. Depth class = $\text{argmax}(\text{probability})$, but the uncertainty field is passed downstream.

Loophole 4 — EI formula weights are arbitrary with no sensitivity analysis The weights (0.4, 0.3, 0.3) in the Extractability Index have no physical justification — they were chosen subjectively. Any ISRO reviewer will immediately ask "why those weights?" Fix: run a sensitivity analysis — compute EI over a grid of weight combinations (e.g., 5×5×5 weight sweep), show that the spatial ranking of top candidate zones is stable (rank correlation > 0.85) across weight choices. This turns a weakness into a strength: "our ranking is robust to weight uncertainty."

Loophole 5 — Dielectric inversion using IEM is not robust at high incidence angles The Integral Equation Model works best at moderate incidence angles (20–50°). DFSAR acquires data at incidence angles up to 70° in polar geometry. At steep angles, IEM gives non-physical ϵ values. Fix: use the Oh (2004) empirical model (validated for high incidence angles) in parallel with IEM. If both agree within 15%, report the IEM value. If they diverge, flag the pixel and use the empirical model as primary.

Loophole 6 — The landing site score couples ice access to a "traversable path" but Block 6 runs before Block 7 The LS_score requires "distance-penalised, EI-weighted average extractable ice value reachable via traversable path," but the rover path planner is Block 7. This is a circular dependency — the path depends on the landing site and the landing site depends on the path. Fix: resolve with a two-pass approach. Pass 1: compute LS_score using Euclidean distance to high-EI zone (approximation). Pass 2: run a lightweight Dijkstra traversal from each candidate site to nearest high-EI zone using the terrain cost raster (without the scientific objective layer), then recompute LS_score using path cost. This breaks the circularity cleanly.

Loophole 7 — NSGA-II on a full DEM grid is computationally intractable in 30 hours NSGA-II on a pixel-level grid graph of a 5m DEM covering a multi-km crater is millions of nodes. It will not converge in a hackathon. Fix: hierarchical planning. Coarse pass (100m LOLA DEM) with NSGA-II → identify Pareto-optimal corridor regions. Fine pass (5m LRO NAC DTM) with A* within those corridors, with the multi-objective cost function baked into a scalar via Chebyshev scalarization of the Pareto-optimal trade-off weights. This runs in minutes and still produces a family of distinct paths.

Loophole 8 — No ground truth validation or confidence reporting Every output map (ice probability, EI, depth class) is produced with no cross-validation against any independent dataset within the same crater. Fix: cross-validate against Mini-RF CPR maps (independent sensor, different wavelength) and LRO LAMP UV albedo (surface frost indicator). Report agreement percentages. Even partial agreement significantly boosts scientific credibility.

Loophole 9 — Agentic AI is mentioned nowhere in v1 The outline is a linear pipeline with no AI collaboration layer. ISRO explicitly asked for "AI-based navigation models." Fix: design a Multi-Agent System (MAS) where each pipeline stage is an autonomous agent with defined inputs, outputs, and inter-agent communication. Agents cross-validate each other's outputs. This is the highest-differentiation addition available.

Loophole 10 — Timeline not specified A 30-hour hackathon with 7 blocks + 3 bonus ideas + MAS is impossible without a timed execution plan. Fix: include a per-hour execution roadmap with hard cutoffs and fallback stubs for each block.

SECTION 1 — PROJECT IDENTITY

Project Name: PRISM — Polarimetric Resource Intelligence System for the Moon

Tagline: "We don't just detect ice — we tell you which ice is worth going to, how to reach it safely, and how much can actually be extracted — driven by a collaborative multi-agent AI system operating entirely on Chandrayaan-2 data."

Core Thesis: A fully data-driven, uncertainty-aware, end-to-end pipeline powered by a Multi-Agent System (MAS) where each analytical domain is handled by a specialized AI agent. Agents communicate, cross-validate, and reach consensus. Every output carries a confidence score. Every number has an uncertainty bound.

SECTION 2 — COMPLETE DATASET MANIFEST

2.1 Primary (Mission-supplied + Mandatory)

Dataset	Resolution	Source	What it gives you
DFSAR L-band SLC	~2m azimuth	pradan.issdc.gov.in/ch2	Deep penetration (3-5m), volume scattering
DFSAR S-band SLC	~2m azimuth	Same portal	Shallow penetration (1-2m), surface sensitivity
OHRC imagery	25 cm/px	Same portal	Surface morphology, boulders, texture

Critical first step on DFSAR: Identify whether data is full-polarimetric (HH, HV, VH, VV) or compact-pol (RH, RV or hybrid). This determines which decomposition algorithm you use. Check the metadata header file (.xml or .hdr) for "polarization_mode" field.

2.2 Terrain

Dataset	Resolution	Source
LOLA DEM	30m and 100m/px	pds-geosciences.wustl.edu
SLDEM2015	60m/px	Same (LOLA + Kaguya merged)
LRO NAC DTM	5m/px (where available)	lroc.sese.asu.edu/data

2.3 Thermal

Dataset	Resolution	Source	Notes
DIVINER bolometric temp grids	240m/px	diviner.ucla.edu and PDS	Annual Tmax and Tmin. Williams et al. 2024 PSJ specifically characterizes Faustini.

2.4 Shadow and Illumination

Dataset	Resolution	Source
ShadowCam (KPLO/Danuri)	1.7m/px	data.im-ldi.com/lunaserv.html
LOLA illumination maps (Mazarico et al. 2011)	250m/px	PDS — fraction time illuminated

2.5 Cross-Validation / Context

Dataset	Source	Use
Mini-RF (LRO) CPR maps	PDS	Independent CPR cross-validation
LRO LAMP UV albedo	PDS	Surface frost proxy
Lunar Prospector neutron H maps	PDS	Background hydrogen context

2.6 Important Data Notes

- DFSAR SLC data comes in complex format (I+Q). You must compute intensity and then derive Stokes parameters before any polarimetric analysis.
- OHRC images may have geometric distortion near poles due to high solar incidence angles — apply orthorectification using the DEM before co-registration.
- DIVINER at 240m/px is coarser than DFSAR. Do NOT apply DIVINER temperature at pixel level to DFSAR maps. Assign DIVINER values by bilinear interpolation and treat them as zone-level priors, not pixel-level ground truth.
- ShadowCam images are from KPLO which was operational as of 2022. Verify coverage for your specific crater region before relying on it.

SECTION 3 — TECHNOLOGY STACK

3.1 Core Language and Scientific Libraries

```
Python 3.10+  
numpy, scipy, pandas  
gdal (osgeo), rasterio, pyproj, shapely  
scikit-learn (Random Forest, cross-validation)  
scikit-image (texture analysis – GLCM)  
matplotlib, seaborn, plotly (visualization)  
pymoo (NSGA-II multi-objective optimization)  
networkx or custom grid graph (path planning)
```

3.2 SAR-Specific

```
SNAP (ESA Sentinel Application Platform) – SAR preprocessing, speckle  
filtering, geocoding  
MIDAS (if supplied by ISRO) – DFSAR-specific processing  
pyroSAR – Python wrapper for SNAP (automates batch SNAP workflows)
```

3.3 Agent Framework (MAS Layer)

```
LangGraph (preferred) – stateful multi-agent DAG orchestration  
OR  
CrewAI – simpler role-based agent collaboration  
OR  
AutoGen (Microsoft) – conversational multi-agent framework  
  
For inter-agent messaging: JSON-serialized confidence-annotated payloads  
LLM backbone: GPT-4o or Claude 3.5 Sonnet (API) for reasoning-heavy agents  
Local computation agents: Pure Python, no LLM needed
```

Architecture Decision: Use LangGraph as the orchestration layer. LangGraph allows directed acyclic graph (DAG) structure between agents with state passing — perfectly matching the PRISM pipeline structure. Each Block = one agent node in the graph.

3.4 GIS and Visualization

```
QGIS (GUI verification and map export)  
ArcGIS Pro (if available)  
Folium or PyDeck (interactive web maps for presentation)
```

3.5 Environment

Conda environment with all above packages
SNAP Toolbox 9.0+ (standalone install)
Docker container (optional) for reproducibility

SECTION 4 — MULTI-AGENT SYSTEM ARCHITECTURE

This is the highest-differentiation element of PRISM. No other team will have this.

4.1 Why MAS?

Each stage of the pipeline requires different expertise: SAR physics, terrain mechanics, thermal modeling, path optimization, geological interpretation. A single monolithic script cannot reason about uncertainty, cross-check between disciplines, or escalate conflicts. A MAS allows:

- Parallel execution of independent analyses
- Cross-agent validation (agent A's output checked by agent B)
- Explicit conflict resolution when agents disagree
- Traceable confidence provenance — every number has a source agent

4.2 Agent Roster

AGENT 1 — PREPROCESSOR PRIME

Domain: Data ingestion, co-registration, quality control **Inputs:** Raw DFSAR SLC, OHRC, LOLA DEM, DIVINER, ShadowCam **Outputs:** Co-registered multiband raster stack; data quality report; polarization mode flag **Key Tasks:**

- Detect full-pol vs compact-pol mode from metadata
- Run multilooking, Refined Lee speckle filtering, radiometric calibration (sigma-nought)
- Terrain correction and geocoding to lunar south polar stereographic (EPSG:104903 or equivalent)
- Co-register all ancillary layers to DFSAR grid using GDAL warp
- Flag pixels affected by layover/shadowing in SAR geometry (these must be excluded from ice analysis)
- Output: GeoTIFF stack + JSON quality report with noise equivalent sigma-zero (NESZ) per band

Best Practice: Compute Equivalent Number of Looks (ENL) after speckle filtering. ENL should be 9–16. Lower ENL means under-filtered (noisy); higher means over-smoothed (resolution lost). Report ENL in the quality report.

Novel Touch: Produce an automatic data quality mask — pixels with low coherence or layover contamination get a quality flag that all downstream agents respect. No downstream agent processes a masked pixel.

AGENT 2 — POLSAR DETECTIVE (Ice Detection)

Domain: Polarimetric analysis, ice probability mapping **Inputs:** Co-registered DFSAR stack from Agent 1; polarization mode flag **Outputs:** Per-pixel ice probability map (0–1); CPR and DOP maps (L and S band); decomposition component maps **Key Tasks:**

Step A — Baseline CPR/DOP:

- $CPR = \sigma^{\circ}_{\text{same}} / \sigma^{\circ}_{\text{cross}}$ (circular polarization ratio)
- DOP = degree of linear polarization from Stokes parameters
- Apply PRL thresholds as baseline: $CPR > 1.0$, $DOP < 0.13$
- This produces a binary ice candidate mask — label it "Level 0 Ice Map"

Step B — Polarimetric Decomposition:

- IF full-polarimetric: Build covariance matrix [C] or coherency matrix [T] per pixel. Apply Yamaguchi 4-component decomposition → P_v (volume), P_d (double bounce), P_s (surface), P_h (helix)
- IF compact-pol (e.g., RH/RV): Reconstruct pseudo-Stokes parameters using Stokes vector approach. Apply m-chi decomposition (Raney 2012) → volume, double-bounce, surface power fractions
- Compute Volume Scattering Fraction ($VSF = P_v / (P_v + P_d + P_s)$). High VSF + high CPR = strong ice indicator. High VSF + low CPR = likely rough dry regolith.

Step C — ML Ice Probability:

- Feature vector per pixel: [CPR_L, CPR_S, DOP_L, DOP_S, σ°_L , σ°_S , VSF, backscatter_ratio_L/S]
- Positive labels: Mini-RF confirmed ice zone polygons (from PRL paper geographic description, NOT CPR values)
- Negative labels: Apollo landing sites (non-PSR, known ice-free), illuminated slopes outside PSR
- Classifier: Random Forest (200 trees, max_depth=12, class_weight='balanced')
- Validation: 5-fold spatial cross-validation on held-out crater sub-regions

- Output: Probability_ice raster (0.0-1.0)

Important Note — Avoid False Positives from Boulders: Boulder fields produce high CPR via double-bounce and create $CPR > 1$ without ice. The VSF ratio resolves this: boulders → high Pd, low VSF. Ice → high Pv, high VSF. If a pixel has $CPR > 1$ but $VSF < 0.3$, flag it as "boulder candidate, not ice" in a separate layer. This is the PRL paper's unresolved ambiguity — you resolve it.

Output layers: CPR_L, CPR_S, DOP_L, DOP_S, VSF, P_ice (probability), Ice_Level0 (binary), BoulderFlag, agent_confidence (= mean prediction probability of RF, averaged over ensemble)

AGENT 3 — DEPTH SOUNDER (Depth Stratification)

Domain: Dual-frequency depth analysis **Inputs:** Ice probability map and CPR maps from Agent 2; Agent 2 confidence field **Outputs:** Depth class map (shallow/deep/none); per-pixel depth probability vector; depth uncertainty raster **Key Tasks:**

Physics Basis:

- L-band (~1.27 GHz) penetration depth in dry lunar regolith $\approx 3-5\text{m}$
- S-band (~3.2 GHz) penetration depth $\approx 1-2\text{m}$
- Penetration depth $\delta = \lambda / (2\pi \times \text{Im}(\sqrt{\epsilon}))$ — varies with regolith dielectric properties

Classification with Uncertainty (fixes Loophole 3): For each pixel where $P_{\text{ice}} > 0.5$:

1. Compute CPR_L uncertainty: $\sigma_{\text{CPR}_L} = \text{CPR}_L \times \sqrt{(2/\text{ENL})}$ (from speckle statistics)
2. Draw 500 Monte Carlo samples of CPR_L and CPR_S from their distributions
3. For each sample, apply thresholds → depth class vote
4. Report: $P(\text{shallow}) = \text{fraction of samples in shallow class}$; $P(\text{deep}) = \text{fraction in deep class}$

Depth classification rules per sample:

- $\text{CPR}_L > 1.0$ AND $\text{CPR}_S > 1.0 \rightarrow \text{shallow (0-2m)}$
- $\text{CPR}_L > 1.0$ AND $\text{CPR}_S \leq 1.0 \rightarrow \text{deep (2-5m)}$
- $\text{CPR}_L \leq 1.0 \rightarrow \text{no detectable ice}$

Final class = argmax(P(shallow), P(deep), P(none)) Uncertainty = $1 - \max(\text{P vector})$ — ranges 0 (certain) to 0.67 (totally uncertain)

Cross-validation with Idea C (Lobate Rim Morphometry): Flag all shallow-zone pixels that overlap the lobate rim region identified in LOLA DEM. If rim morphometry (Block C)

predicts ice at 1–2m depth in the same region → agreement event. Log and report.

AGENT 4 — THERMO GUARDIAN (Thermal and Stability Analysis)

Domain: Thermal stability, volatile preservation timescales **Inputs:** DIVINER temperature grids; shadow maps; depth map from Agent 3 **Outputs:** Thermal Stability Score (TS) raster; Cold Trap Classification map; Volatile Age Estimate (Idea A) **Key Tasks:**

Step A — Thermal Stability Score: Interpolate DIVINER 240m/px T_{max} to DFSAR pixel grid (bilinear interpolation; flag pixels where DEM and DIVINER elevation are inconsistent by >100m — these are mis-registrations, not real anomalies).

Stability thresholds:

- $T_{\max} < 70\text{K}$: TS = 1.0
- $70 \leq T_{\max} < 90\text{K}$: TS = 0.7
- $90 \leq T_{\max} < 110\text{K}$: TS = 0.3
- $T_{\max} \geq 110\text{K}$: TS = 0.0

Important: The doubly shadowed crater interior is expected to be $\leq 55\text{K}$ based on DIVINER+Williams 2024 — so the entire floor will be TS = 1.0. However, the crater rim transition zone and approach corridor to the crater will vary. This is crucial for landing site and rover path planning. Do not skip rim pixel characterization.

Step B — Cold Trap Classification: Combine LOLA illumination fraction + DIVINER T_{max}:

- PSR (illumination = 0%) + $T_{\max} < 110\text{K}$ = Cold Trap (volatile-preserving)
- PSR + $T_{\max} < 70\text{K}$ = Extreme Cold Trap
- Doubly shadowed sub-crater within PSR + $T_{\max} < 55\text{K}$ = Super Cold Trap (target zone)

Step C — Volatile Age Estimation (Idea A integration): From ShadowCam imagery (1.7m/px): count craters overlapping the doubly shadowed crater floor using automated crater detection (CDA via circular Hough transform or DeepMoon CNN). Apply Neukum Production Function (NPF) crater chronology for lunar south pole. Output: surface age estimate in Gy with 1-sigma uncertainty. Interpret: older surface → longer ice preservation → higher confidence in stable long-term deposit. Tag this as a "geological confidence bonus" in the final ISRU ranking.

Best Practice: Do not attempt to count craters < 50m diameter from LOLA DEM. Use ShadowCam for small crater detection. Count only craters clearly within the doubly shadowed region, not overlapping the rim.

AGENT 5 — VOLUME ORACLE (Ice Volume Estimation)

Domain: Dielectric inversion, volumetric ice fraction, Monte Carlo uncertainty **Inputs:**

Calibrated σ^0 maps from Agent 1; ice probability from Agent 2; depth map from Agent 3; TS from Agent 4 **Outputs:** Dielectric constant raster; volumetric ice fraction raster; total ice volume (median + 90% CI); extractable ice volume **Key Tasks:**

Step A — Dielectric Inversion:

- Primary method: Integral Equation Model (IEM) inversion
 - Retrieve ϵ from σ^0 using IEM forward model (Fung et al. 1992)
 - Valid incidence range: 20°–50°. Flag pixels outside this range.
- Backup method: Oh (2004) empirical model (valid 10°–70° incidence)
 - Uses σ^0_{hh} and σ^0_{hv} to retrieve ϵ and surface RMS height simultaneously
- Consistency check: $|\epsilon_{IEM} - \epsilon_{Oh}| < 0.5 \rightarrow$ use IEM. Larger divergence $\rightarrow$ flag pixel, use Oh model as primary.

Reference Values:

- Dry lunar regolith: $\epsilon_{real} \approx 2.7\text{--}3.1$ (Apollo samples; Carrier et al. 1991)
- Water ice: $\epsilon_{real} \approx 3.15\text{--}3.17$ at cryogenic temperatures
- Ice/regolith mixture: ϵ_{mix} computed via Polder–van Santen mixing rule

Step B — Mixing Model: Polder–van Santen: $\epsilon_{eff} = \epsilon_{host} + f \times (\epsilon_{ice} - \epsilon_{host}) \times (\epsilon_{ice} + 2 \times \epsilon_{host}) / (3 \times \epsilon_{host})$ — solve for f (volumetric ice fraction). Constraint: $0 \leq f \leq 0.5$ (lunar ice fractions above 50% are geologically implausible; clip and flag outliers).

Step C — Volume Calculation: Per pixel: $V_{pixel} = pixel_area \times layer_depth \times f$ Layer depth from Agent 3: 2.0m (shallow zone), 5.0m (deep zone), 0 (no-ice zone)

Step D — Monte Carlo Uncertainty (1000 runs): Vary: f ($\pm \epsilon$ inversion uncertainty $\rightarrow$ propagated via $\delta f / \delta \epsilon$); regolith bulk density (1400–1900 kg/m³ from Apollo variance); penetration depth ($\pm 15\%$ of nominal) Output: V_{median} , $V_{5th_percentile}$, $V_{95th_percentile}$ (90% CI) Report: "Total ice volume in F2 target zone = X million tonnes (90% CI: Y–Z million tonnes)"

Novel Addition: Report ice water equivalent (IWE) in tonnes, not just cubic meters. Convert using ice density = 917 kg/m³. This is the currency ISRU planners think in. "X million litres of water" is a mission headline.

AGENT 6 — TERRAIN SCOUT (Morphology and Safety)

Domain: Terrain safety, slope analysis, boulder detection, landing site scoring **Inputs:** LOLA DEM (30m and 5m); OHRC imagery; EI map from coordination (post-Agent 7 first pass); ShadowCam; illumination maps **Outputs:** Slope raster; roughness raster; boulder density map; solar illumination raster; Landing Site Score (LS_score); top 3 candidate sites with ranked justification table **Key Tasks:**

Step A — Terrain Metrics:

- Slope: compute from DEM using Horn's method (rasterio/GDAL built-in). Flag pixels with slope $> 15^\circ$ as no-go (lander stability limit).
- Surface roughness: RMS height deviation over 5×5 pixel window on 5m DEM.
- Boulder detection: OHRC texture analysis using GLCM (Gray-Level Co-occurrence Matrix). Features: contrast, entropy, energy, homogeneity. High contrast + high entropy = high boulder density.
- Crater rim identification: Circular Hough transform on OHRC gradient image. Flag rim pixels — steep transition zones, risky for rovers.

Step B — Solar Illumination: From LOLA illumination fraction map (Mazarico 2011): per pixel, fraction of time with solar illumination.

- $f_{\text{sun}} > 0.5$: "illuminated ridge" (potential solar charging waypoint)
- $f_{\text{sun}} = 0.0$: permanent shadow (PSR) — rover needs battery reserve to traverse
- $0 < f_{\text{sun}} < 0.5$: "twilight zone" — partial illumination, use carefully

Step C — Landing Site Score (Two-Pass, fixes Loophole 6): Pass 1 (Euclidean approximation): $\text{LS_score_1} = 0.35 \times \text{ice_access_euclidean} + 0.25 \times \text{slope_score} + 0.20 \times \text{solar_score} + 0.15 \times \text{dist_to_psr} + 0.05 \times \text{comm_score}$

Pass 2 (Path-cost weighted — after Agent 7 provides terrain cost raster):

- Run Dijkstra from each candidate landing site to nearest high-EI zone ($\text{EI} > 0.7$)
- $\text{path_cost} = \text{sum of terrain cost raster values along shortest path}$
- $\text{ice_access_path} = \text{EI_target} / (1 + \text{path_cost_normalized})$
- $\text{LS_score_2} = 0.35 \times \text{ice_access_path} + 0.25 \times \text{slope_score} + 0.20 \times \text{solar_score} + 0.15 \times \text{traverse_cost_normalized} + 0.05 \times \text{comm_score}$ Final $\text{LS_score} = \text{LS_score_2}$

Output: Top 3 ranked candidate landing sites with full attribute table: (slope, roughness, solar fraction, distance to high-EI zone, traverse path cost, LS_score, justification).

Novel Touch — Downrange Visibility: Check line-of-sight from each candidate landing site to Earth (using LOLA DEM profile — does the crater rim block the direct link at low

Earth elevation angles?). Even a 10% reduction in communication windows affects science return. Penalize accordingly.

AGENT 7 — ISRU ARCHITECT (Extractability Index + MAS Coordinator)

Domain: Extractability scoring, sensitivity analysis, multi-agent consensus integration, ISRU priority map **Inputs:** Depth map (Agent 3); TS raster (Agent 4); ice volume (Agent 5); terrain metrics (Agent 6); ice probability (Agent 2) **Outputs:** Extractability Index (EI) raster; ISRU Priority Map; Extractable Ice Volume; Sensitivity Analysis Report; MAS Consensus Report **Key Tasks:**

Step A — EI Computation:

Component 1 — Depth Accessibility (DA):

- $P_{\text{shallow}} > 0.6$: DA = 1.0
- $P_{\text{deep}} > 0.6$: DA = 0.4
- $P_{\text{none}} > 0.6$: DA = 0.0
- Uncertain (max $P < 0.6$): DA = 0.2 (partial credit, flag for review)

Component 2 — Regolith Compaction Score (RC):

- RMSH from LOLA DEM: normalize to [0,1] (min=0cm, max=20cm per Apollo regolith variability)
- GLCM texture entropy from OHRC: normalize to [0,1]
- $RC = 0.6 \times \text{roughness_norm} + 0.4 \times (1 - \text{boulder_entropy_norm})$

Component 3 — Thermal Stability Score (TS): from Agent 4

Final EI = $w_1 \times DA + w_2 \times RC + w_3 \times TS$ (baseline: $w_1=0.4, w_2=0.3, w_3=0.3$)

Step B — Sensitivity Analysis (fixes Loophole 4):

- Generate 125 weight combinations: $w_1 \in \{0.3, 0.35, 0.4, 0.45, 0.5\}$, $w_2 \in \text{same}$, $w_3 = 1 - w_1 - w_2$ (only valid combinations where all weights > 0.1)
- For each combination, compute EI map, extract top 20% EI pixels, compute Spearman rank correlation vs baseline
- Report: mean rank correlation across weight sets. If $\rho > 0.85 \rightarrow$ "ranking is robust"
- Visualize: sensitivity heatmap of top EI zone stability

Step C — Extractable Volume: $V_{\text{extractable}} = \sum (V_{\text{pixel}} \times EI_{\text{pixel}})$ across all pixels where $P_{\text{ice}} > 0.5$ Report with 90% CI from Agent 5's Monte Carlo.

Step D — MAS Consensus Integration: This is the coordination function. Agent 7 collects confidence reports from all agents and checks for inter-agent conflicts:

Conflict Check 1 (Agent 2 vs Agent 5): Does the ice probability map agree spatially with elevated dielectric constant zones? Expect >70% overlap. If < 50% → flag discrepancy, run a dedicated conflict resolution pass.

Conflict Check 2 (Agent 3 vs Idea C morphometry): Do radar-derived depth zones agree with morphometric depth estimates from lobate rim analysis? If depth class disagrees by more than one zone → log as "unresolved depth uncertainty" and widen CI.

Conflict Check 3 (Agent 4 vs Agent 6): Are all high-EI pixels in thermally stable zones (TS > 0.7)? Any high-EI pixel in a TS = 0.0 zone → contradiction; recalculate EI with TS = 0 override.

Output: Consensus Report JSON with per-agent confidence scores and inter-agent agreement metrics. This is the MAS's unique contribution to the final result.

AGENT 8 — NAVIGATOR (Rover Traverse Planning)

Domain: Multi-objective path optimization, energy budgeting, charging waypoints **Inputs:** EI map (Agent 7); terrain cost raster (Agent 6); solar illumination map; landing site candidates; DEM **Outputs:** Pareto-optimal path family; energy budget per path; recommended path with waypoints; 3D traverse visualization **Key Tasks:**

Step A — Cost Raster Construction: Three layers (all normalized to [0,1]):

- C_{terrain} : slope penalty $e^{(\text{slope_deg}/10)}$, multiplied by $(1 + \text{boulder_density})$
- C_{solar} : $(1 - f_{\text{sun}}) \rightarrow \text{shadow} = \text{max penalty}$. PSR pixel = 1.0 (must traverse on battery reserves)
- C_{science} : $(1 - \text{EI}) \rightarrow \text{path toward high-EI zones is rewarded (lower cost)}$

Step B — Hierarchical Path Planning (fixes Loophole 7): Level 1 — Coarse planning (100m LOLA DEM):

- Run NSGA-II (pymoo) on a ~200×200 node graph
- Objectives: minimize ($C_{\text{terrain_sum}}$, $C_{\text{solar_sum}}$, $C_{\text{science_sum}}$)
- Output: Pareto front of non-dominated solutions (typically 20–50 paths)
- Select 3 representative paths: min terrain cost, min solar cost, balanced (closest to ideal Chebyshev point)
- These define corridor regions (50m buffer around each Pareto path)

Level 2 — Fine refinement (5m LRO NAC DTM if available, else 30m LOLA):

- Within each corridor, run A* with scalarized cost: $C_{\text{total}} = \alpha \times C_{\text{terrain}} + \beta \times C_{\text{solar}} + \gamma \times C_{\text{science}}$
- Use (α, β, γ) from the 3 representative Pareto points
- Output: 3 fine-grained paths with per-waypoint coordinates

Step C — Charging Waypoints: Between landing site and crater entry, identify all pixels where $f_{\text{sun}} > 0.5$. Among these, find locally accessible "ridge peaks" (local maximum illumination within 500m). Insert these as mandatory rest waypoints for battery recharge. Compute: energy gained per stop = solar_panel_area $\times$ efficiency $\times$ mean_illumination $\times$ dwell_time. Dwell time = energy_needed / power_input.

Step D — Energy Budget: For each path:

- Energy consumed = Σ (rolling_resistance $\times$ rover_mass $\times$ g_moon $\times$ segment_length) for each segment + instrument power $\times$ total_time
- Energy available = solar_panel_area $\times$ efficiency $\times \Sigma$ ($f_{\text{sun_per_segment}} \times$ travel_time_per_segment) + battery_capacity_from_last_charge
- Flag path as "infeasible" if energy_available $<$ energy_consumed with $< 20\%$ margin
- Rover parameters: mass = 25kg (Pragyan-class), solar panel = 50W (at 1AU), battery = 100 Wh, rolling resistance coefficient = 0.15 (lunar regolith), lunar g = 1.62 m/s²

Step E — Output:

- 3D visualization of terrain + all 3 paths overlaid (matplotlib 3D or plotly)
- Table: path_id, total_length_m, terrain_risk_score, solar_feasibility (%), science_score_EI_sum, energy_margin (%)
- Recommended path = balanced Pareto path with positive energy margin

SECTION 5 — MAS COMMUNICATION PROTOCOL

5.1 Agent Message Format

Every inter-agent message is a JSON payload:

```
json
```

```

{
  "sender_agent": "POLARSAR_DETECTIVE",
  "recipient_agent": "DEPTH_SOUNDER",
  "timestamp": "2025-06-17T04:00:00Z",
  "payload_type": "raster_reference",
  "payload": {
    "file": "outputs/P_ice_map.tif",
    "crs": "EPSG:104903",
    "resolution_m": 4.5,
    "agent_confidence": 0.81,
    "quality_mask": "outputs/quality_mask.tif",
    "notes": "Compact-pol mode detected; m-chi decomposition used instead of Ya
  }
}

```

5.2 Orchestration with LangGraph

```

START → PREPROCESSOR_PRIME
→ [POLARSAR_DETECTIVE, THERMO_GUARDIAN, TERRAIN_SCOUT] (parallel)
→ DEPTH_SOUNDER (waits for POLARSAR output)
→ VOLUME_ORACLE (waits for DEPTH_SOUNDER and POLARSAR)
→ ISRU_ARCHITECT (waits for DEPTH_SOUNDER, VOLUME_ORACLE, THERMO_GUARDIAN,
TERRAIN_SCOUT)
→ TERRAIN_SCOUT pass 2 (waits for ISRU_ARCHITECT EI map)
→ NAVIGATOR (waits for pass 2 TERRAIN_SCOUT and ISRU_ARCHITECT)
→ CONSENSUS_REPORT → END

```

5.3 Conflict Resolution Protocol

- Level 1 (Minor disagreement, <20% spatial mismatch): Log discrepancy, proceed with higher-confidence agent output
- Level 2 (Major disagreement, 20-50% spatial mismatch): Both agents re-examine a 3×3 pixel neighborhood for context; update outputs; ISRU_ARCHITECT applies 0.5× confidence weight to disputed pixels
- Level 3 (Fundamental contradiction, >50% mismatch or physical impossibility): Escalate to human review flag; ISRU_ARCHITECT excludes disputed region from extractable volume estimate with explicit note

SECTION 6 — THREE NOVEL CONTRIBUTIONS (IDEAS A, B, C)

Idea A — Volatile Stability Age Estimation

Agent: THERMO GUARDIAN (integrated in Step C) **Method:** Crater size-frequency distribution on doubly shadowed crater floor from ShadowCam **Tool:** DeepMoon CNN or Circular Hough Transform on ShadowCam gradient images **Output:** Surface age in Gyr with uncertainty; interpretive confidence label

Important: Only count craters > 50m diameter (smaller ones approach ShadowCam resolution limits and are unreliable for chronometry). The NPF isochrons for polar regions have higher uncertainty than equatorial regions — report error bars accordingly (typically $\pm 30\text{--}40\%$).

Idea B — Multi-Crater Comparative ISRU Ranking

When to do it: In parallel with main pipeline during the coarse-pass stages (hours 18–22)

Craters: F2 (main target), F3, H3, S1 (from PRL paper) **Method:** Run PRISM pipeline at 100m LOLA resolution only (no fine-grained SAR analysis for comparison craters — too time-consuming) for F3, H3, S1 **Output:** Table with 6 columns: Crater | Ice Probability (mean P_{ice}) | Dominant Depth Zone | EI (median) | Extractable Volume (order of magnitude) | Landing Accessibility Score | Overall ISRU Rank **Claim:** "For Chandrayaan-4 ISRU mission planning, optimal target priority order is $F2 > [X] > [Y] > [Z]$, for the following quantified reasons."

Idea C — Lobate Rim Morphometry as Ice Depth Proxy

Agent: Standalone analysis (can be run by any team member with LOLA DEM and ShadowCam) **Method:**

1. Identify lobate rim in LOLA DEM: rim radius R , rim height H , ejecta blanket extent L
2. Compute rim asymmetry index: $\sigma(H \text{ along azimuth}) / \text{mean}(H)$
3. Compare R/D (rim radius to crater diameter ratio) against ice-excitation models (Marco Figuera et al. 2022)
4. Back-calculate ice depth at impact using the scaling relationship from Fassett et al. 2022
5. Cross-validate: does morphometric depth (e.g., 1.5m) agree with Agent 3 shallow/deep class at the same location? **Output:** Independent ice depth estimate with model uncertainty. Agreement with Agent 3 → "cross-validated confirmation." Disagreement → "unresolved — additional data needed" (also valid scientific output).

SECTION 7 — CROSS-VALIDATION FRAMEWORK (Fixes Loophole 8)

This is the scientific credibility layer no other team will have.

Validation Pair	Method	Expected Agreement	What if Disagrees
DFSAR CPR vs Mini-RF CPR	Pixel-level spatial correlation (Pearson r)	$r > 0.65$ (different wavelengths, some disagreement expected)	Report disagreement range, note wavelength sensitivity difference
Ice probability map vs LRO LAMP UV albedo	Overlap fraction in high- P_{ice} zone vs high-albedo zone	$> 50\%$ overlap	Flag spatial offset, note albedo = surface frost only, CPR = subsurface
Depth class (Agent 3) vs lobate rim morphometry (Idea C)	Class match at rim intersection pixels	$\geq 70\%$ agreement	Widen CI, report as "evidence" not "confirmation"
EI map vs illumination map	Anti-correlation check (high EI should be in PSR, low illumination)	Spearman $\rho < -0.5$	Alert: EI may include illuminated zones — check thermal stability override
Volume estimate vs published estimates (if any)	Order of magnitude comparison	Within 1 order of magnitude	Examine penetration depth assumption

SECTION 8 — 30-HOUR EXECUTION ROADMAP

Hour	Milestone	Who	Fallback if behind
0-1	Dataset download and inventory; verify DFSAR polarization mode	All	If PRADAN slow, use backup Mini-RF data from PDS to begin CPR baseline
1-4	Agent 1 (PREPROCESSOR PRIME): SNAP processing, co-registration	1-2 people	Skip terrain correction; do it post-analysis on outputs
4-7	Agent 2 (POLARSAR DETECTIVE): CPR/DOP maps, decomposition, Level-0 ice map	1-2 people	Skip RF classifier; use CPR threshold only as Level-0 (still valid)

Hour	Milestone	Who	Fallback if behind
7-10	Agent 3 (DEPTH SOUNDER): depth stratification, Monte Carlo depth probabilities	1 person	Skip Monte Carlo; use hard threshold depth map
7-10	Agent 4 (THERMO GUARDIAN): DIVINER processing, TS raster, cold trap map	1 person (parallel)	Use published Tmax values for Faustini from Williams 2024 directly
10-13	Agent 5 (VOLUME ORACLE): dielectric inversion, f, MC volume estimate	1 person	Use simplified $\varepsilon = f \times \varepsilon_{\text{ice}} + (1-f) \times \varepsilon_{\text{regolith}}$ (CRIM model) instead of IEM
10-13	Agent 6 (TERRAIN SCOUT) Pass 1: slope, roughness, boulder density, LS Pass 1	1 person (parallel)	Use 30m LOLA only; skip OHRC texture
13-16	Agent 7 (ISRU ARCHITECT): EI map, sensitivity analysis, consensus integration	1-2 people	Skip sensitivity analysis; use baseline weights only
16-18	Agent 6 Pass 2: EI-informed landing site score; finalize top 3 sites	1 person	Use Pass 1 LS_score
16-18	Idea C (Lobate Rim Morphometry)	1 person (parallel)	Use qualitative description if quantitative model not available
18-22	Agent 8 (NAVIGATOR): coarse NSGA-II + fine A*, energy budget	1-2 people	Use A* only with scalar cost = $0.5 \times \text{terrain} + 0.3 \times \text{solar} + 0.2 \times (1 - \text{EI})$
18-22	Idea B (Multi-crater ranking) at coarse resolution	1 person (parallel)	Rank using CPR + slope only as proxy EI
22-24	Idea A (Volatile age estimation)	1 person	Skip if ShadowCam coverage not confirmed; note as "pending"
24-27	MAS consensus report; cross-validation checks; final maps	All	Prioritize cross-validation over bonus ideas
27-30	Presentation assembly; key result figures; narrative rehearsal	All	—

Non-negotiable by Hour 20: Ice probability map, depth class map, EI map, at least one valid landing site, at least one traversable path. Everything else is a bonus.

SECTION 9 — KEY FIGURES TO PRODUCE FOR PRESENTATION

1. **Multi-band composite:** RGB = [CPR_L, CPR_S, VSF] overlaid on ShadowCam base image. High-ice zones appear distinctive color.
 2. **Ice Probability Map:** Continuous 0–1 colormap (viridis), threshold contour at 0.5 and 0.7.
 3. **Depth Stratification Map:** Three-class color map (shallow = blue, deep = orange, none = grey) with uncertainty overlay (transparency proportional to depth uncertainty).
 4. **ISRU Priority Map (EI):** Green (EI>0.7) / Yellow (0.4–0.7) / Red (<0.4) — with agent confidence as hatching.
 5. **Landing Site Candidates:** Hillshade DEM + top 3 site markers + LS_score shown as label. Approach corridor shown.
 6. **Rover Traverse Plot:** 3 Pareto paths overlaid on terrain + EI map. Charging waypoints marked. Energy margin shown as path width.
 7. **MAS Architecture Diagram:** Agent nodes, arrows showing data flow, conflict events shown as orange markers.
 8. **Cross-Validation Summary:** Table or radar chart showing agreement scores across all 5 validation pairs.
 9. **Volume Uncertainty:** Box plot or violin plot showing Monte Carlo distribution of total ice volume and extractable ice volume side by side.
 10. **Pareto Front Plot:** 2D scatter of (terrain cost, science reward) for all Pareto paths; 3 representative paths highlighted.
-

SECTION 10 — THE ONE-SENTENCE WIN

"PRISM is the first multi-agent AI system to operationalize Chandrayaan-2 DFSAR data into a fully uncertainty-quantified, cross-validated ISRU mission blueprint — resolving the ambiguity between ice and boulders, stratifying ice by extractable depth, and navigating toward the highest-value ice with a solar-constrained rover — not by human selection, but by collaborative machine intelligence derived entirely from Indian space mission data."

APPENDIX — IMPORTANT REFERENCES TO CITE

- PRL 2026 paper on DFSAR ice detection at Faustini (the baseline you extend)
- Williams et al. 2024 PSJ — DIVINER characterization of Faustini thermal environment
- Yamaguchi et al. 2005 — 4-component polarimetric decomposition
- Raney 2012 — m-chi decomposition for compact-pol SAR
- Oh 2004 — Empirical model for dielectric retrieval at high incidence angles
- Fung et al. 1992 — Integral Equation Model (IEM)
- Polder & van Santen 1946 — Dielectric mixing model
- Mazarico et al. 2011 — LOLA illumination maps of lunar south pole
- Marco Figuera et al. 2022 — Ice-excavation crater morphology scaling
- Fassett et al. 2022 — Crater chronology for polar regions
- Apollo regolith properties — Carrier et al. 1991 (bulk density, dielectric values)